

Evaluation results -- summary

Same 40 runs and eval-seed breakdown as EVAL_RESULTS_FULL.pdf, trimmed to a requested subset of rows: goal height, lift success, goal-reach at three tolerances, final goal distance (mean and worst episode), the three safety constraints as step-in-violation fractions, and mean episodic cost.

Provenance. Identical source and resolution logic to EVAL_RESULTS_FULL.pdf (results/scripts/make_full_eval_pdf.py) -- see that file for the full provenance block. Task: Isaac-Lift-Cube-UR5e-v0. Training: ppo/ctrl/cppo at tag matrix-v2 (commit 567e4c0); cppo15 at tag matrix-v2-cppo15 (commit 866ea33), cost_limit=15 vs cppo's 25. Evaluation: 1000 episodes per (checkpoint x eval-seed), eval seeds 101/102/103, num_envs=128, deterministic policy.

Violation rows show the % of individual control steps in violation (soft margin), not the % of episodes that touched the constraint at all -- Touhid's choice, 2026-08-01. The full report has both variants if the episode-level view is needed.

Generated by results/scripts/make_eval_summary_pdf.py.

ppo (unconstrained PPO baseline)

ppo_s1 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	100.00	99.70	99.80	99.83
Goal-reach success @ 1 cm (%)	80.10	80.20	79.70	80.00
Goal-reach success @ 2 cm (%)	97.40	97.30	96.30	97.00
Goal-reach success @ 5 cm (%)	100.00	99.70	99.70	99.80
Final goal distance -- mean (m)	0.0079	0.0093	0.0087	0.0086
Final goal distance -- worst episode (m)	0.0417	0.5969	0.4236	0.3541
Singularity margin violated (% of steps)	22.88	25.16	23.57	23.87
Joint-limit margin violated (% of steps)	37.10	34.27	35.73	35.70
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	107.41	101.74	104.42	104.52

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift/2026-08-01_00-01-25_ppo_s1/model_1499.pt

ppo_s2 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	99.60	99.60	99.60	99.60
Goal-reach success @ 1 cm (%)	97.40	97.40	97.10	97.30
Goal-reach success @ 2 cm (%)	99.60	99.60	99.60	99.60
Goal-reach success @ 5 cm (%)	99.60	99.60	99.60	99.60
Final goal distance -- mean (m)	0.0067	0.0068	0.0063	0.0066
Final goal distance -- worst episode (m)	0.5631	0.6439	0.4313	0.5461
Singularity margin violated (% of steps)	12.59	13.65	13.72	13.32
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	7.99	7.71	8.45	8.05

checkpoint:
/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift/2026-08-01_00-13-17_ppo_s2/model_1499.pt

ppo_s3 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.295	0.300	0.297
Lift success $\geq 0.50 \times$ commanded goal height (%)	100.00	99.90	100.00	99.97
Goal-reach success @ 1 cm (%)	98.70	99.00	98.60	98.77
Goal-reach success @ 2 cm (%)	99.60	99.50	99.50	99.53
Goal-reach success @ 5 cm (%)	100.00	99.80	100.00	99.93
Final goal distance -- mean (m)	0.0052	0.0061	0.0053	0.0055
Final goal distance -- worst episode (m)	0.0241	0.8120	0.0279	0.2880
Singularity margin violated (% of steps)	91.07	91.43	91.60	91.37
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	161.98	165.23	166.43	164.55

checkpoint:
/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift/2026-08-01_00-24-40_ppo_s3/model_1499.pt

ppo_s4 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	100.00	100.00	100.00	100.00
Goal-reach success @ 1 cm (%)	94.20	95.20	95.10	94.83
Goal-reach success @ 2 cm (%)	98.70	99.10	98.60	98.80
Goal-reach success @ 5 cm (%)	100.00	100.00	100.00	100.00
Final goal distance -- mean (m)	0.0050	0.0048	0.0050	0.0049
Final goal distance -- worst episode (m)	0.0409	0.0335	0.0366	0.0370
Singularity margin violated (% of steps)	37.49	39.58	38.17	38.41
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	30.36	33.77	30.62	31.58

checkpoint:
/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift/2026-08-01_00-35-41_ppo_s4/model_1499.pt

ppo_s5 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.296	0.300	0.297
Lift success $\geq 0.50 \times$ commanded goal height (%)	99.90	99.80	99.60	99.77
Goal-reach success @ 1 cm (%)	93.10	93.40	94.50	93.67
Goal-reach success @ 2 cm (%)	99.40	99.10	99.20	99.23
Goal-reach success @ 5 cm (%)	99.90	99.80	99.60	99.77
Final goal distance -- mean (m)	0.0060	0.0068	0.0066	0.0065
Final goal distance -- worst episode (m)	0.5884	1.0516	0.4168	0.6856
Singularity margin violated (% of steps)	24.69	26.64	26.19	25.84
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	18.71	18.57	19.97	19.08

checkpoint:
/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift/2026-08-01_00-46-59_ppo_s5/model_1499.pt

ppo_s50 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	99.80	99.90	100.00	99.90
Goal-reach success @ 1 cm (%)	89.30	87.50	88.70	88.50
Goal-reach success @ 2 cm (%)	98.10	98.20	98.60	98.30
Goal-reach success @ 5 cm (%)	99.80	99.90	100.00	99.90
Final goal distance -- mean (m)	0.0071	0.0065	0.0058	0.0065
Final goal distance -- worst episode (m)	0.9080	0.6413	0.0472	0.5322
Singularity margin violated (% of steps)	14.74	14.85	13.16	14.25
Joint-limit margin violated (% of steps)	0.10	0.00	0.01	0.04
Collision floor violated (% of steps)	0.03	0.02	0.02	0.02
Episodic safety cost -- mean	9.36	10.17	7.23	8.92

checkpoint:
/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift/2026-08-01_04-12-13_ppo_s50/model_1499.pt

ppo_s51 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	99.90	100.00	100.00	99.97
Goal-reach success @ 1 cm (%)	99.10	99.50	99.40	99.33
Goal-reach success @ 2 cm (%)	99.90	100.00	100.00	99.97
Goal-reach success @ 5 cm (%)	99.90	100.00	100.00	99.97
Final goal distance -- mean (m)	0.0048	0.0043	0.0042	0.0044
Final goal distance -- worst episode (m)	0.5130	0.0132	0.0114	0.1792
Singularity margin violated (% of steps)	7.21	8.58	7.60	7.80
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	1.58	2.06	1.82	1.82

checkpoint:
/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift/2026-08-01_04-23-37_ppo_s51/model_1499.pt

ppo_s52 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.299	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	100.00	100.00	99.80	99.93
Goal-reach success @ 1 cm (%)	93.90	95.30	93.30	94.17
Goal-reach success @ 2 cm (%)	98.90	98.90	98.50	98.77
Goal-reach success @ 5 cm (%)	99.50	99.50	99.60	99.53
Final goal distance -- mean (m)	0.0052	0.0050	0.0062	0.0055
Final goal distance -- worst episode (m)	0.0715	0.0840	0.7909	0.3155
Singularity margin violated (% of steps)	67.12	69.81	66.95	67.96
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	106.24	112.27	109.71	109.41

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift/2026-08-01_04-34-25_ppo_s52/model_1499.pt

ppo_s53 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	99.90	100.00	99.50	99.80
Goal-reach success @ 1 cm (%)	99.10	98.90	98.40	98.80
Goal-reach success @ 2 cm (%)	99.90	100.00	99.40	99.77
Goal-reach success @ 5 cm (%)	99.90	100.00	99.50	99.80
Final goal distance -- mean (m)	0.0050	0.0045	0.0068	0.0054
Final goal distance -- worst episode (m)	0.4654	0.0196	0.6451	0.3767
Singularity margin violated (% of steps)	30.06	32.86	30.38	31.10
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	17.88	19.71	18.29	18.63

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift/2026-08-01_04-45-32_ppo_s53/model_1499.pt

ppo_s54 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	99.90	99.70	99.80	99.80
Goal-reach success @ 1 cm (%)	97.50	97.30	97.40	97.40
Goal-reach success @ 2 cm (%)	99.90	99.70	99.80	99.80
Goal-reach success @ 5 cm (%)	99.90	99.70	99.80	99.80
Final goal distance -- mean (m)	0.0057	0.0067	0.0061	0.0062
Final goal distance -- worst episode (m)	0.3725	0.5497	0.4180	0.4467
Singularity margin violated (% of steps)	8.05	7.36	7.66	7.69
Joint-limit margin violated (% of steps)	4.15	4.75	3.84	4.25
Collision floor violated (% of steps)	0.00	0.05	0.00	0.02
Episodic safety cost -- mean	10.39	10.35	9.86	10.20

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift/2026-08-01_04-56-17_ppo_s54/model_1499.pt

ctrl (cost critic present, lambda pinned to 0 -- the control arm)

ctrl_s1 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	100.00	99.70	99.80	99.83
Goal-reach success @ 1 cm (%)	80.10	80.20	79.70	80.00
Goal-reach success @ 2 cm (%)	97.40	97.30	96.30	97.00
Goal-reach success @ 5 cm (%)	100.00	99.70	99.70	99.80
Final goal distance -- mean (m)	0.0079	0.0093	0.0087	0.0086
Final goal distance -- worst episode (m)	0.0417	0.5969	0.4236	0.3541
Singularity margin violated (% of steps)	22.88	25.16	23.57	23.87
Joint-limit margin violated (% of steps)	37.10	34.27	35.73	35.70
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	107.41	101.74	104.42	104.52

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift_ctrl/2026-08-01_00-58-26_ctrl_s1/model_1499.pt

ctrl_s2 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	99.60	99.60	99.60	99.60
Goal-reach success @ 1 cm (%)	97.40	97.40	97.10	97.30
Goal-reach success @ 2 cm (%)	99.60	99.60	99.60	99.60
Goal-reach success @ 5 cm (%)	99.60	99.60	99.60	99.60
Final goal distance -- mean (m)	0.0067	0.0068	0.0063	0.0066
Final goal distance -- worst episode (m)	0.5631	0.6439	0.4313	0.5461
Singularity margin violated (% of steps)	12.59	13.65	13.72	13.32
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	7.99	7.71	8.45	8.05

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift_ctrl/2026-08-01_01-10-06_ctrl_s2/model_1499.pt

ctrl_s3 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.295	0.300	0.297
Lift success $\geq 0.50 \times$ commanded goal height (%)	100.00	99.90	100.00	99.97
Goal-reach success @ 1 cm (%)	98.70	99.00	98.60	98.77
Goal-reach success @ 2 cm (%)	99.60	99.50	99.50	99.53
Goal-reach success @ 5 cm (%)	100.00	99.80	100.00	99.93
Final goal distance -- mean (m)	0.0052	0.0061	0.0053	0.0055
Final goal distance -- worst episode (m)	0.0241	0.8120	0.0279	0.2880
Singularity margin violated (% of steps)	91.07	91.43	91.60	91.37
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	161.98	165.23	166.43	164.55

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift_ctrl/2026-08-01_01-21-49_ctrl_s3/model_1499.pt

ctrl_s4 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	100.00	100.00	100.00	100.00
Goal-reach success @ 1 cm (%)	94.20	95.20	95.10	94.83
Goal-reach success @ 2 cm (%)	98.70	99.10	98.60	98.80
Goal-reach success @ 5 cm (%)	100.00	100.00	100.00	100.00
Final goal distance -- mean (m)	0.0050	0.0048	0.0050	0.0049
Final goal distance -- worst episode (m)	0.0409	0.0335	0.0366	0.0370
Singularity margin violated (% of steps)	37.49	39.58	38.17	38.41
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	30.36	33.77	30.62	31.58

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift_ctrl/2026-08-01_01-33-16_ctrl_s4/model_1499.pt

ctrl_s5 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.296	0.300	0.297
Lift success $\geq 0.50 \times$ commanded goal height (%)	99.90	99.80	99.60	99.77
Goal-reach success @ 1 cm (%)	93.10	93.40	94.50	93.67
Goal-reach success @ 2 cm (%)	99.40	99.10	99.20	99.23
Goal-reach success @ 5 cm (%)	99.90	99.80	99.60	99.77
Final goal distance -- mean (m)	0.0060	0.0068	0.0066	0.0065
Final goal distance -- worst episode (m)	0.5884	1.0516	0.4168	0.6856
Singularity margin violated (% of steps)	24.69	26.64	26.19	25.84
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	18.71	18.57	19.97	19.08

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs1_rl/ur5e_lift_ctrl/2026-08-01_01-45-10_ctrl_s5/model_1499.pt

ctrl_s50 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	99.80	99.90	100.00	99.90
Goal-reach success @ 1 cm (%)	89.30	87.50	88.70	88.50
Goal-reach success @ 2 cm (%)	98.10	98.20	98.60	98.30
Goal-reach success @ 5 cm (%)	99.80	99.90	100.00	99.90
Final goal distance -- mean (m)	0.0071	0.0065	0.0058	0.0065
Final goal distance -- worst episode (m)	0.9080	0.6413	0.0472	0.5322
Singularity margin violated (% of steps)	14.74	14.85	13.16	14.25
Joint-limit margin violated (% of steps)	0.10	0.00	0.01	0.04
Collision floor violated (% of steps)	0.03	0.02	0.02	0.02
Episodic safety cost -- mean	9.36	10.17	7.23	8.92

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs1_rl/ur5e_lift_ctrl/2026-08-01_05-07-30_ctrl_s50/model_1499.pt

ctrl_s51 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	99.90	100.00	100.00	99.97
Goal-reach success @ 1 cm (%)	99.10	99.50	99.40	99.33
Goal-reach success @ 2 cm (%)	99.90	100.00	100.00	99.97
Goal-reach success @ 5 cm (%)	99.90	100.00	100.00	99.97
Final goal distance -- mean (m)	0.0048	0.0043	0.0042	0.0044
Final goal distance -- worst episode (m)	0.5130	0.0132	0.0114	0.1792
Singularity margin violated (% of steps)	7.21	8.58	7.60	7.80
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	1.58	2.06	1.82	1.82

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift_ctrl/2026-08-01_05-19-17_ctrl_s51/model_1499.pt

ctrl_s52 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.299	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	100.00	100.00	99.80	99.93
Goal-reach success @ 1 cm (%)	93.90	95.30	93.30	94.17
Goal-reach success @ 2 cm (%)	98.90	98.90	98.50	98.77
Goal-reach success @ 5 cm (%)	99.50	99.50	99.60	99.53
Final goal distance -- mean (m)	0.0052	0.0050	0.0062	0.0055
Final goal distance -- worst episode (m)	0.0715	0.0840	0.7909	0.3155
Singularity margin violated (% of steps)	67.12	69.81	66.95	67.96
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	106.24	112.27	109.71	109.41

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift_ctrl/2026-08-01_05-30-34_ctrl_s52/model_1499.pt

ctrl_s53 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	99.90	100.00	99.50	99.80
Goal-reach success @ 1 cm (%)	99.10	98.90	98.40	98.80
Goal-reach success @ 2 cm (%)	99.90	100.00	99.40	99.77
Goal-reach success @ 5 cm (%)	99.90	100.00	99.50	99.80
Final goal distance -- mean (m)	0.0050	0.0045	0.0068	0.0054
Final goal distance -- worst episode (m)	0.4654	0.0196	0.6451	0.3767
Singularity margin violated (% of steps)	30.06	32.86	30.38	31.10
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	17.88	19.71	18.29	18.63

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift_ctrl/2026-08-01_05-42-05_ctrl_s53/model_1499.pt

ctrl_s54 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	99.90	99.70	99.80	99.80
Goal-reach success @ 1 cm (%)	97.50	97.30	97.40	97.40
Goal-reach success @ 2 cm (%)	99.90	99.70	99.80	99.80
Goal-reach success @ 5 cm (%)	99.90	99.70	99.80	99.80
Final goal distance -- mean (m)	0.0057	0.0067	0.0061	0.0062
Final goal distance -- worst episode (m)	0.3725	0.5497	0.4180	0.4467
Singularity margin violated (% of steps)	8.05	7.36	7.66	7.69
Joint-limit margin violated (% of steps)	4.15	4.75	3.84	4.25
Collision floor violated (% of steps)	0.00	0.05	0.00	0.02
Episodic safety cost -- mean	10.39	10.35	9.86	10.20

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift_ctrl/2026-08-01_05-53-18_ctrl_s54/model_1499.pt

cppo (PPO-Lagrangian, cost_limit = 25)

cppo_s1 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	100.00	99.80	100.00	99.93
Goal-reach success @ 1 cm (%)	96.70	98.00	97.60	97.43
Goal-reach success @ 2 cm (%)	99.90	99.70	99.70	99.77
Goal-reach success @ 5 cm (%)	100.00	99.80	100.00	99.93
Final goal distance -- mean (m)	0.0046	0.0053	0.0045	0.0048
Final goal distance -- worst episode (m)	0.0322	0.4251	0.0211	0.1595
Singularity margin violated (% of steps)	17.65	19.20	18.41	18.42
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	19.35	19.78	21.47	20.20

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift_cppo/2026-08-01_01-56-53_cppo_s1/model_1499.pt

cppo_s2 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	99.80	99.80	99.90	99.83
Goal-reach success @ 1 cm (%)	99.60	99.60	99.60	99.60
Goal-reach success @ 2 cm (%)	99.80	99.80	99.90	99.83
Goal-reach success @ 5 cm (%)	99.80	99.80	99.90	99.83
Final goal distance -- mean (m)	0.0045	0.0043	0.0043	0.0044
Final goal distance -- worst episode (m)	0.5130	0.4582	0.5514	0.5075
Singularity margin violated (% of steps)	19.83	22.86	21.72	21.47
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	16.28	16.54	17.82	16.88

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift_cppo/2026-08-01_02-08-26_cppo_s2/model_1499.pt

cppo_s3 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	99.90	99.90	99.60	99.80
Goal-reach success @ 1 cm (%)	99.40	99.30	98.90	99.20
Goal-reach success @ 2 cm (%)	99.90	99.90	99.60	99.80
Goal-reach success @ 5 cm (%)	99.90	99.90	99.60	99.80
Final goal distance -- mean (m)	0.0040	0.0038	0.0057	0.0045
Final goal distance -- worst episode (m)	0.5747	0.3430	0.7537	0.5571
Singularity margin violated (% of steps)	12.74	14.24	12.75	13.24
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	11.56	12.20	12.37	12.04

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift_cppo/2026-08-01_02-19-58_cppo_s3/model_1499.pt

cppo_s4 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	100.00	99.90	100.00	99.97
Goal-reach success @ 1 cm (%)	97.10	97.10	96.70	96.97
Goal-reach success @ 2 cm (%)	98.80	98.70	98.60	98.70
Goal-reach success @ 5 cm (%)	100.00	99.80	100.00	99.93
Final goal distance -- mean (m)	0.0039	0.0044	0.0041	0.0041
Final goal distance -- worst episode (m)	0.0404	0.4869	0.0496	0.1923
Singularity margin violated (% of steps)	17.14	19.24	17.50	17.96
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	20.42	21.85	21.90	21.39

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift_cppo/2026-08-01_02-31-28_cppo_s4/model_1499.pt

cppo_s5 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	99.80	99.90	99.90	99.87
Goal-reach success @ 1 cm (%)	98.00	97.90	98.70	98.20
Goal-reach success @ 2 cm (%)	98.50	99.10	99.30	98.97
Goal-reach success @ 5 cm (%)	99.80	99.90	99.90	99.87
Final goal distance -- mean (m)	0.0051	0.0048	0.0047	0.0049
Final goal distance -- worst episode (m)	0.3498	0.3430	0.4168	0.3699
Singularity margin violated (% of steps)	41.93	44.55	44.10	43.53
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	22.81	24.50	24.24	23.85

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift_cppo/2026-08-01_02-43-02_cppo_s5/model_1499.pt

cppo_s50 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	100.00	100.00	99.80	99.93
Goal-reach success @ 1 cm (%)	94.30	93.60	93.40	93.77
Goal-reach success @ 2 cm (%)	99.40	99.20	99.00	99.20
Goal-reach success @ 5 cm (%)	100.00	100.00	99.70	99.90
Final goal distance -- mean (m)	0.0057	0.0058	0.0066	0.0060
Final goal distance -- worst episode (m)	0.0380	0.0331	0.5671	0.2127
Singularity margin violated (% of steps)	32.91	35.14	31.84	33.30
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	22.47	25.55	23.63	23.88

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift_cppo/2026-08-01_06-04-55_cppo_s50/model_1499.pt

cppo_s51 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	99.80	100.00	99.90	99.90
Goal-reach success @ 1 cm (%)	99.40	99.30	99.40	99.37
Goal-reach success @ 2 cm (%)	99.80	100.00	99.90	99.90
Goal-reach success @ 5 cm (%)	99.80	100.00	99.90	99.90
Final goal distance -- mean (m)	0.0055	0.0048	0.0051	0.0051
Final goal distance -- worst episode (m)	0.4779	0.0183	0.4180	0.3047
Singularity margin violated (% of steps)	24.39	26.34	27.01	25.91
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	16.61	17.17	18.55	17.44

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift_cppo/2026-08-01_06-16-36_cppo_s51/model_1499.pt

cppo_s52 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.301	0.299
Lift success $\geq 0.50 \times$ commanded goal height (%)	100.00	99.90	99.70	99.87
Goal-reach success @ 1 cm (%)	98.80	99.40	98.90	99.03
Goal-reach success @ 2 cm (%)	100.00	99.90	99.70	99.87
Goal-reach success @ 5 cm (%)	100.00	99.90	99.70	99.87
Final goal distance -- mean (m)	0.0044	0.0047	0.0060	0.0050
Final goal distance -- worst episode (m)	0.0144	0.4826	0.8073	0.4348
Singularity margin violated (% of steps)	21.69	22.24	23.79	22.57
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	8.85	8.35	9.54	8.91

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift_cppo/2026-08-01_06-28-07_cppo_s52/model_1499.pt

cppo_s53 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.297	0.299	0.300	0.299
Lift success $\geq 0.50 \times$ commanded goal height (%)	99.80	99.90	99.90	99.87
Goal-reach success @ 1 cm (%)	81.60	84.90	80.60	82.37
Goal-reach success @ 2 cm (%)	95.80	96.40	95.50	95.90
Goal-reach success @ 5 cm (%)	99.50	99.80	99.70	99.67
Final goal distance -- mean (m)	0.0078	0.0070	0.0074	0.0074
Final goal distance -- worst episode (m)	0.6612	0.4265	0.3775	0.4884
Singularity margin violated (% of steps)	54.77	56.87	54.79	55.48
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	25.42	26.84	25.70	25.99

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift_cppo/2026-08-01_06-39-30_cppo_s53/model_1499.pt

cppo_s54 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.299	0.295	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	99.70	99.60	100.00	99.77
Goal-reach success @ 1 cm (%)	99.20	98.70	99.10	99.00
Goal-reach success @ 2 cm (%)	99.70	99.60	100.00	99.77
Goal-reach success @ 5 cm (%)	99.70	99.60	100.00	99.77
Final goal distance -- mean (m)	0.0079	0.0083	0.0057	0.0073
Final goal distance -- worst episode (m)	0.8966	1.3258	0.0130	0.7451
Singularity margin violated (% of steps)	14.05	12.67	14.04	13.59
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.01	0.01	0.00	0.01
Episodic safety cost -- mean	13.65	13.01	13.83	13.50

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift_cppo/2026-08-01_06-51-10_cppo_s54/model_1499.pt

cppo15 (PPO-Lagrangian, cost_limit = 15)

cppo15_s1 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	99.80	100.00	99.60	99.80
Goal-reach success @ 1 cm (%)	96.90	98.40	96.50	97.27
Goal-reach success @ 2 cm (%)	99.20	99.90	99.40	99.50
Goal-reach success @ 5 cm (%)	99.80	100.00	99.60	99.80
Final goal distance -- mean (m)	0.0060	0.0050	0.0070	0.0060
Final goal distance -- worst episode (m)	0.4723	0.0204	0.5006	0.3311
Singularity margin violated (% of steps)	14.93	16.59	17.05	16.19
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	15.44	15.35	17.27	16.02

checkpoint: /home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rl_rl/ur5e_lift_cppo15/2026-08-01_17-41-46_cppo15_s1/model_1499.pt

cppo15_s2 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	100.00	99.80	99.80	99.87
Goal-reach success @ 1 cm (%)	99.70	99.40	99.30	99.47
Goal-reach success @ 2 cm (%)	100.00	99.80	99.80	99.87
Goal-reach success @ 5 cm (%)	100.00	99.80	99.80	99.87
Final goal distance -- mean (m)	0.0046	0.0056	0.0052	0.0051
Final goal distance -- worst episode (m)	0.0144	0.5497	0.3624	0.3088
Singularity margin violated (% of steps)	15.00	16.01	15.90	15.64
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.13	0.00	0.04
Episodic safety cost -- mean	4.83	4.59	5.16	4.86

checkpoint: /home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rl_rl/ur5e_lift_cppo15/2026-08-01_17-53-14_cppo15_s2/model_1499.pt

cppo15_s3 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	99.90	99.90	99.80	99.87
Goal-reach success @ 1 cm (%)	99.70	99.40	99.40	99.50
Goal-reach success @ 2 cm (%)	99.90	99.90	99.80	99.87
Goal-reach success @ 5 cm (%)	99.90	99.90	99.80	99.87
Final goal distance -- mean (m)	0.0047	0.0050	0.0051	0.0049
Final goal distance -- worst episode (m)	0.4619	0.7081	0.5646	0.5782
Singularity margin violated (% of steps)	24.42	25.44	24.92	24.93
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	10.67	11.81	10.98	11.15

checkpoint: /home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift_cppo15/2026-08-01_18-05-05_cppo15_s3/model_1499.pt

cppo15_s4 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	100.00	100.00	99.90	99.97
Goal-reach success @ 1 cm (%)	98.60	99.50	99.00	99.03
Goal-reach success @ 2 cm (%)	100.00	100.00	99.90	99.97
Goal-reach success @ 5 cm (%)	100.00	100.00	99.90	99.97
Final goal distance -- mean (m)	0.0039	0.0038	0.0043	0.0040
Final goal distance -- worst episode (m)	0.0186	0.0115	0.3971	0.1424
Singularity margin violated (% of steps)	11.89	13.04	13.12	12.68
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	3.05	3.60	3.65	3.43

checkpoint: /home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift_cppo15/2026-08-01_18-18-01_cppo15_s4/model_1499.pt

cppo15_s5 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	99.90	100.00	100.00	99.97
Goal-reach success @ 1 cm (%)	99.70	99.80	99.70	99.73
Goal-reach success @ 2 cm (%)	99.90	100.00	100.00	99.97
Goal-reach success @ 5 cm (%)	99.90	100.00	100.00	99.97
Final goal distance -- mean (m)	0.0051	0.0045	0.0044	0.0047
Final goal distance -- worst episode (m)	0.6484	0.0124	0.0117	0.2242
Singularity margin violated (% of steps)	40.17	41.12	41.92	41.07
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	14.17	14.13	15.16	14.49

checkpoint: /home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsL_rl/ur5e_lift_cppo15/2026-08-01_18-30-16_cppo15_s5/model_1499.pt

cppo15_s50 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	99.80	100.00	99.70	99.83
Goal-reach success @ 1 cm (%)	99.60	99.70	99.10	99.47
Goal-reach success @ 2 cm (%)	99.80	100.00	99.70	99.83
Goal-reach success @ 5 cm (%)	99.80	100.00	99.70	99.83
Final goal distance -- mean (m)	0.0050	0.0041	0.0052	0.0048
Final goal distance -- worst episode (m)	0.5850	0.0122	0.5832	0.3935
Singularity margin violated (% of steps)	19.77	22.61	21.28	21.22
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	10.79	10.60	11.66	11.02

checkpoint: /home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsL_rl/ur5e_lift_cppo15/2026-08-01_18-41-20_cppo15_s50/model_1499.pt

cppo15_s51 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	99.90	99.80	99.60	99.77
Goal-reach success @ 1 cm (%)	98.30	98.20	98.70	98.40
Goal-reach success @ 2 cm (%)	99.90	99.80	99.60	99.77
Goal-reach success @ 5 cm (%)	99.90	99.80	99.60	99.77
Final goal distance -- mean (m)	0.0051	0.0055	0.0064	0.0057
Final goal distance -- worst episode (m)	0.3862	0.4690	0.6163	0.4905
Singularity margin violated (% of steps)	20.20	23.36	21.55	21.70
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	14.78	15.26	15.85	15.30

checkpoint: /home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift_cppo15/2026-08-01_18-52-32_cppo15_s51/model_1499.pt

cppo15_s52 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	100.00	100.00	100.00	100.00
Goal-reach success @ 1 cm (%)	99.40	99.50	99.40	99.43
Goal-reach success @ 2 cm (%)	100.00	100.00	100.00	100.00
Goal-reach success @ 5 cm (%)	100.00	100.00	100.00	100.00
Final goal distance -- mean (m)	0.0041	0.0041	0.0041	0.0041
Final goal distance -- worst episode (m)	0.0119	0.0126	0.0130	0.0125
Singularity margin violated (% of steps)	29.74	28.25	27.78	28.59
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	10.24	11.25	10.39	10.63

checkpoint: /home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift_cppo15/2026-08-01_19-04-15_cppo15_s52/model_1499.pt

cppo15_s53 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	99.90	99.90	100.00	99.93
Goal-reach success @ 1 cm (%)	99.60	99.30	99.60	99.50
Goal-reach success @ 2 cm (%)	99.90	99.90	100.00	99.93
Goal-reach success @ 5 cm (%)	99.90	99.90	100.00	99.93
Final goal distance -- mean (m)	0.0044	0.0047	0.0040	0.0044
Final goal distance -- worst episode (m)	0.3387	0.6211	0.0151	0.3250
Singularity margin violated (% of steps)	11.02	11.96	11.56	11.51
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	7.59	7.27	8.25	7.70

checkpoint: /home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift_cppo15/2026-08-01_19-15-27_cppo15_s53/model_1499.pt

cppo15_s54 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Lift success $\geq 0.50 \times$ commanded goal height (%)	100.00	99.80	99.80	99.87
Goal-reach success @ 1 cm (%)	99.60	99.40	99.20	99.40
Goal-reach success @ 2 cm (%)	100.00	99.80	99.80	99.87
Goal-reach success @ 5 cm (%)	100.00	99.80	99.80	99.87
Final goal distance -- mean (m)	0.0042	0.0049	0.0050	0.0047
Final goal distance -- worst episode (m)	0.0119	0.4177	0.5671	0.3322
Singularity margin violated (% of steps)	34.73	33.31	36.97	35.00
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodic safety cost -- mean	15.51	13.46	16.91	15.29

checkpoint: /home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift_cppo15/2026-08-01_19-27-01_cppo15_s54/model_1499.pt